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M.Tech in Information Technology

Design Thinking Lab (Full Stack MERN Lab)

(MIT415DL)

“Campus Find”

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Introduction

- Lost and found item management is a common challenge in large institutions like campuses, offices, and public facilities.
- Items such as ID cards, wallets, phones, and documents are frequently misplaced.
- Traditional methods (notice boards, manual registers) are inefficient and lead to low recovery rates.
- This project develops a web-based system using MERN stack to address these issues.
- Design Thinking ensures user-centric design; MERN chosen for scalability, JavaScript consistency, and rapid development.

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Existing system

- There is no web-based Lost and Found system for the campus.
- Results in frustration, time loss, and reduced recovery chances for users.

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Identification of Problem

- Empathize phase: Conducted surveys and interviews with students, faculty, and security personnel.
- Key insights: Lack of awareness on reporting, difficulty tracking items, delays due to unstructured processes.
- Observations: Overcrowded notice boards, incomplete records, no standardized handover.

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Literature Review

- Reviewed MERN stack resources: MongoDB for flexible NoSQL, Express/Node.js for APIs, React for dynamic UIs.
- Design Thinking principles from Tim Brown (Harvard Business Review, 2008).
- Software engineering practices from Ian Sommerville (10th ed., 2016).
- RESTful architecture from Roy Fielding's dissertation (2000).
- No specific literature on campus lost and found systems in report; focused on full-stack web technologies.

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Problem Definition

- Core Problem: Absence of centralized, efficient platform for managing lost and found items in campuses.
- User Pain Points: Difficulty reporting losses, no real-time visibility, physical visits to multiple locations.
- Real-World Importance: Reduces financial/emotional loss, improves efficiency and trust in institutions.
- Scope: Includes user auth, item reporting/search/matching, claims, admin moderation; excludes AI image recognition, mobile apps.
- Target Users: Students, faculty, staff, admins in campus environment.

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Objectives

- Build a centralized web-based platform for reporting and managing lost and found items.
- Implement secure authentication with role-based access for users, staff, and administrators.
- Enable detailed item reporting with efficient search and filtering.
- Provide secure claim processing.
- Support administrative moderation, item lifecycle tracking, and scalable backend storage.

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Methodology

- Design Thinking Framework: Empathize (surveys, personas, empathy maps, observations).
- Define (problem statement, user requirements, POV statements, scope).
- Ideate (feature brainstorming, innovation mapping, prioritization).
- Prototype (UI mockups, partial MERN features).
- Test (user feedback, usability validation, improvements like clearer labels and navigation).
- Iterative approach ensured alignment with user needs.

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Proposed Tools and Techniques

- Technology Stack: MERN (MongoDB for NoSQL database, Express.js/Node.js for backend APIs, React for frontend UI).
- Tools: Visual Studio Code (editor), Git/GitHub (version control),

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Expected Outcome

- Fully functional full-stack application integrating frontend, backend, and database for end-to-end workflows.
- User-validated solution: Intuitive, effective in addressing traditional challenges.
- Improved usability: Faster searches, real-time visibility, reduced time for matches.
- Enhanced performance: Centralized repository, transparency, higher recovery rates.
- Foundation for enhancements like image-based matching and mobile integration.

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Conclusion

- Summary: Solved inefficient lost and found management by providing a structured web-based system for reporting, searching, and claiming items.
- Value Delivered: Simplifies processes, real-time updates, secure verification; benefits users with better recovery and admins with streamlined workflows.
- Future Scope: Image-based matching with computer vision, mobile apps (Android/iOS), email/SMS notifications, SSO integration, analytics dashboards.

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Thank-you

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